Dataset Validation

Surgical Scene Semantic Segmentation

Dataset: sarrarp50

Number of classes: 10

Complete-dataset distribution analysis

This report evaluates class presence, pixel-level distribution, split consistency, and qualitative segmentation annotations.

All quantitative statistics are computed using the complete samples available in each split.

1. Dataset Overview

Dataset: sarrarp50

Split	Number of samples
val	3252
train	13043

Classes

```
0  —  Background
1  —  Tool clasper
2  —  Tool wrist
3  —  Tool shaft
4  —  Suturing needle
5  —  Thread
6  —  Suction tool
7  —  Needle Holder
8  —  Clamps
9  —  Catheter
```

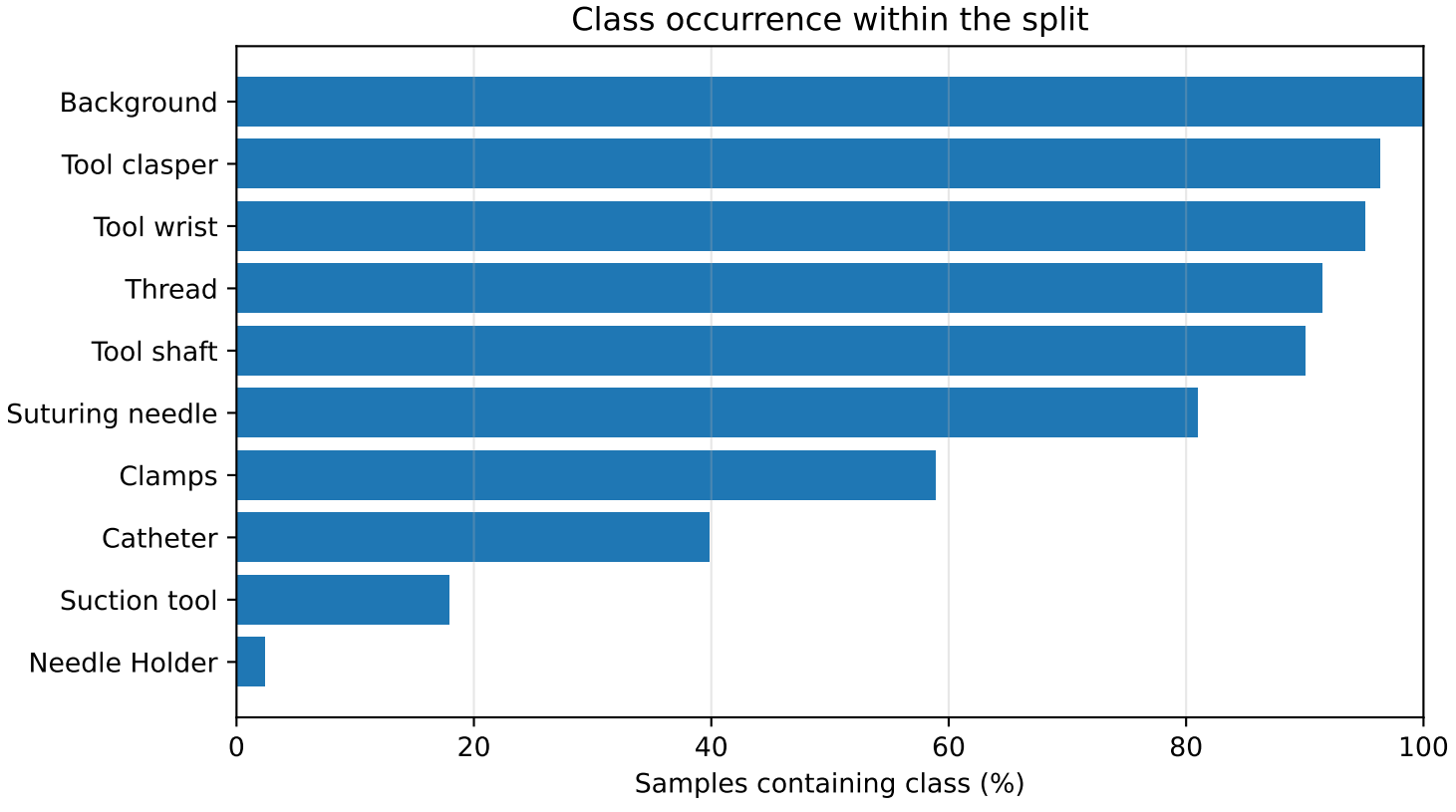
Methodology

The validation procedure evaluates every sample in each split. No sampling or maximum-sample limit is applied to the quantitative analysis.

2. Class Presence

Split: val

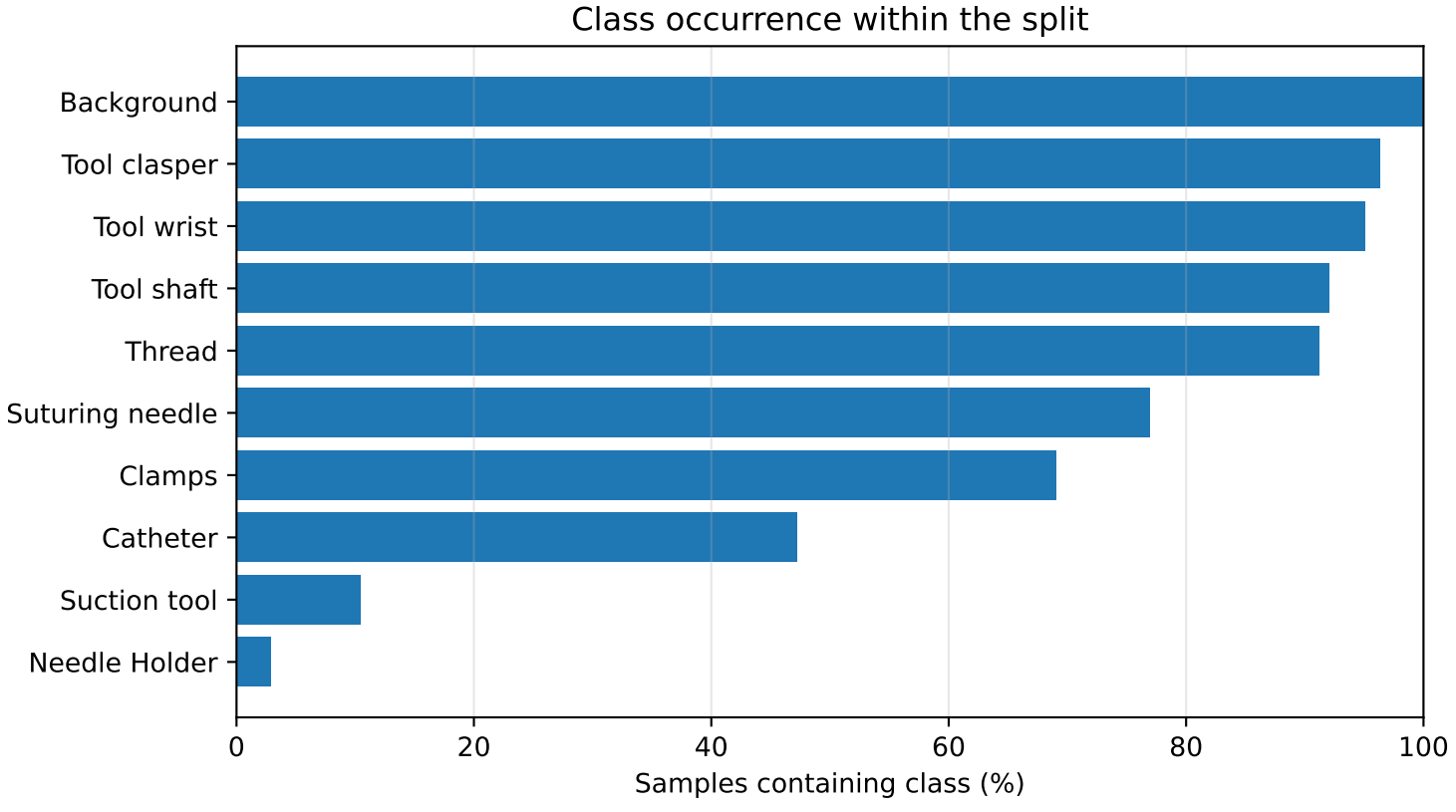
Class	Samples	Samples containing class
Background	3,252	100.00%
Tool clasper	3,134	96.37%
Tool wrist	3,093	95.11%
Tool shaft	2,929	90.07%
Suturing needle	2,633	80.97%
Thread	2,976	91.51%
Suction tool	583	17.93%
Needle Holder	77	2.37%
Clamps	1,916	58.92%
Catheter	1,296	39.85%



2. Class Presence

Split: train

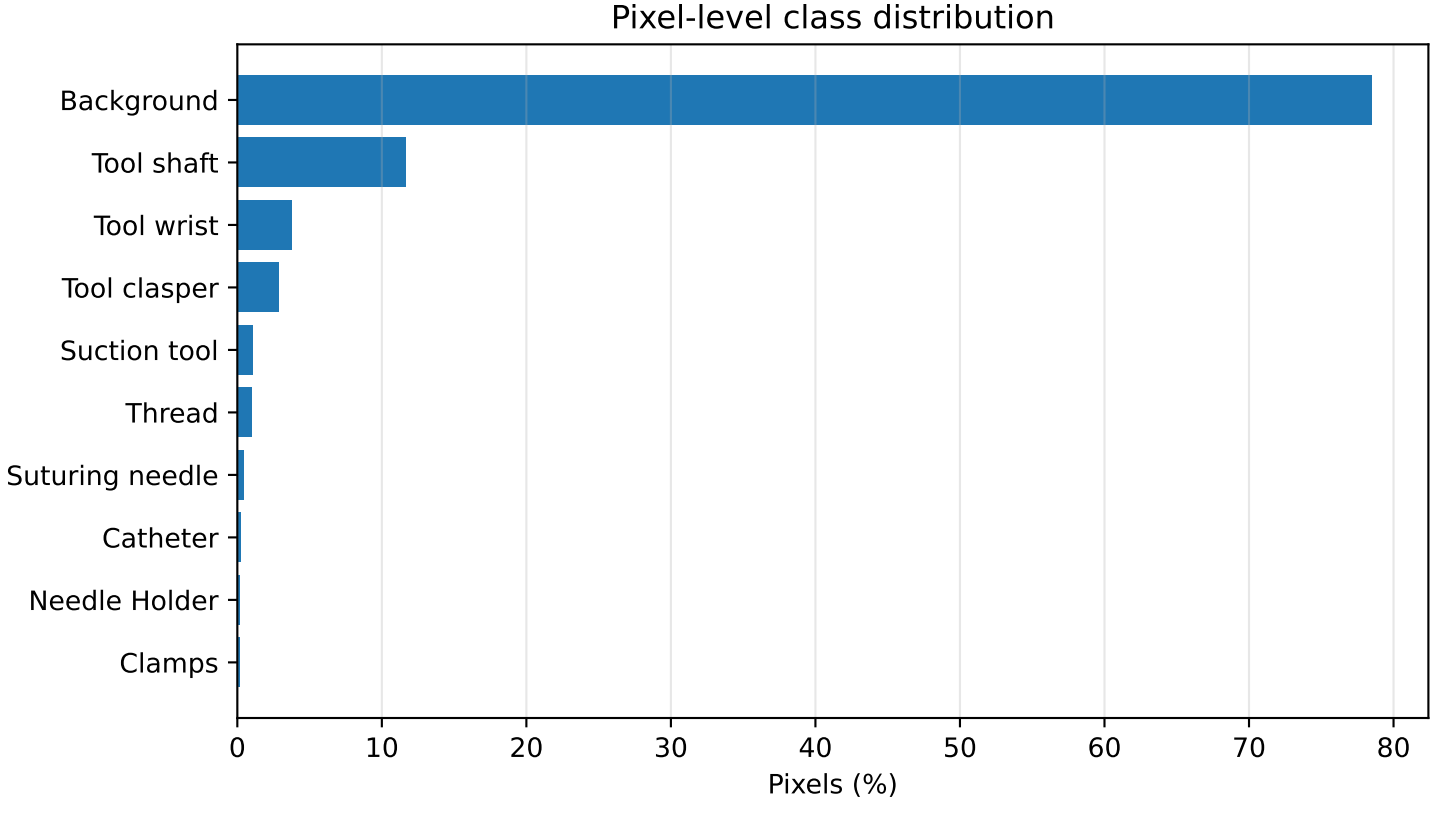
Class	Samples	Samples containing class
Background	13,043	100.00%
Tool clasper	12,567	96.35%
Tool wrist	12,401	95.08%
Tool shaft	12,003	92.03%
Suturing needle	10,037	76.95%
Thread	11,898	91.22%
Suction tool	1,359	10.42%
Needle Holder	381	2.92%
Clamps	9,011	69.09%
Catheter	6,162	47.24%



3. Pixel Distribution

Split: val

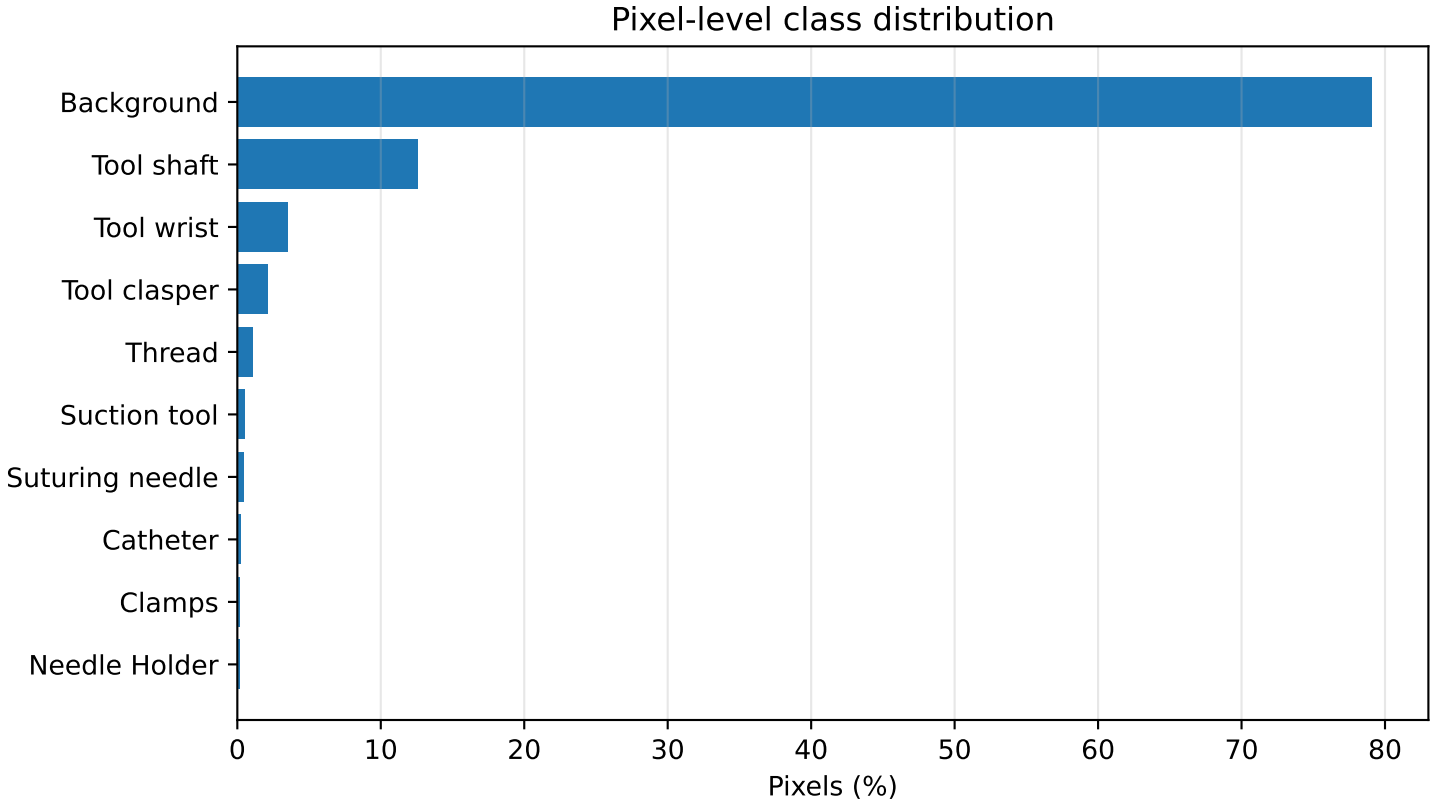
Class	Pixels	Pixel distribution
Background	5,292,652,647	78.4870%
Tool clasper	194,966,049	2.8912%
Tool wrist	253,584,299	3.7605%
Tool shaft	788,227,767	11.6890%
Suturing needle	32,265,352	0.4785%
Thread	68,096,173	1.0098%
Suction tool	74,001,677	1.0974%
Needle Holder	13,283,449	0.1970%
Clamps	11,157,312	0.1655%
Catheter	15,112,475	0.2241%

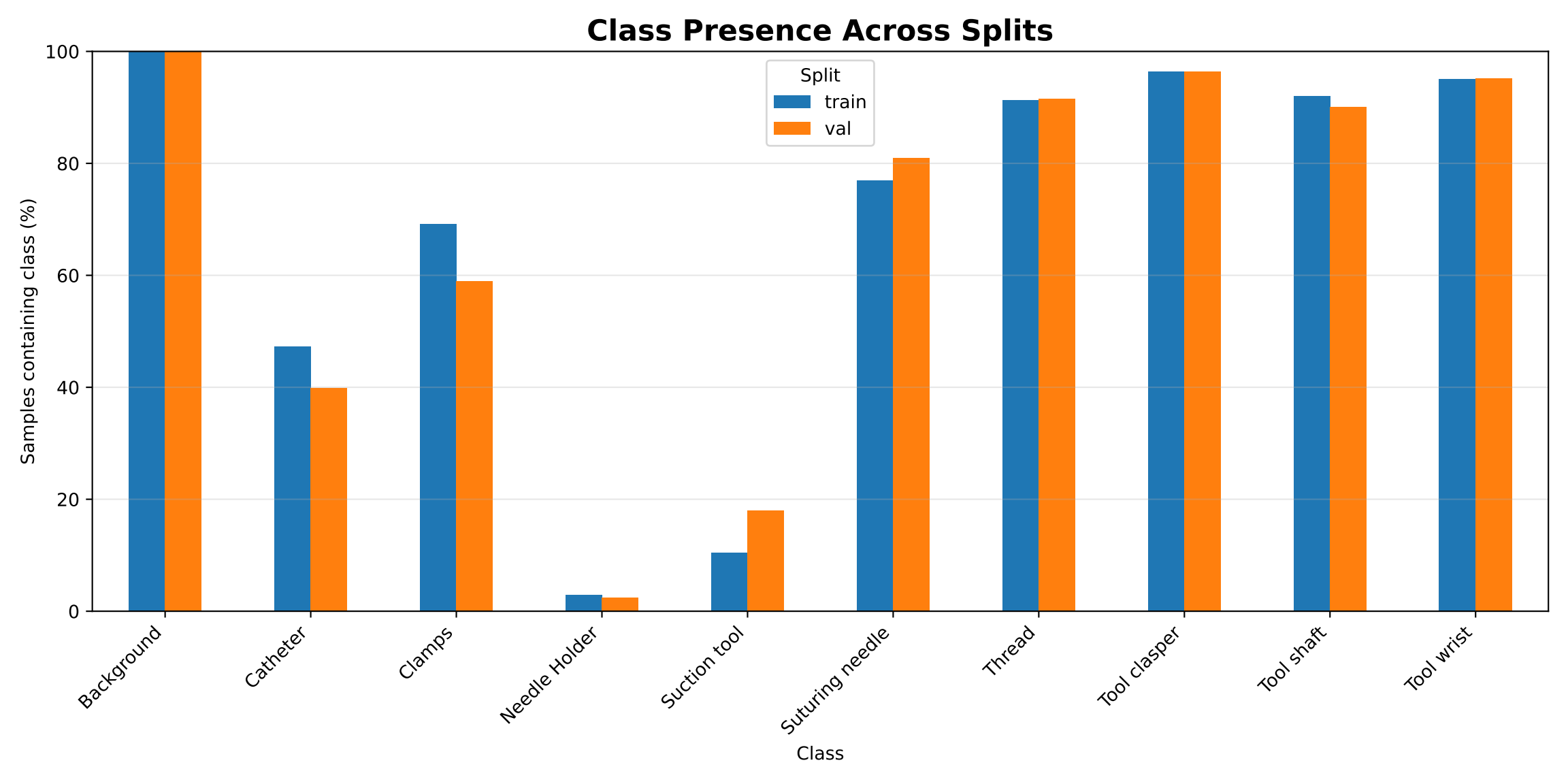


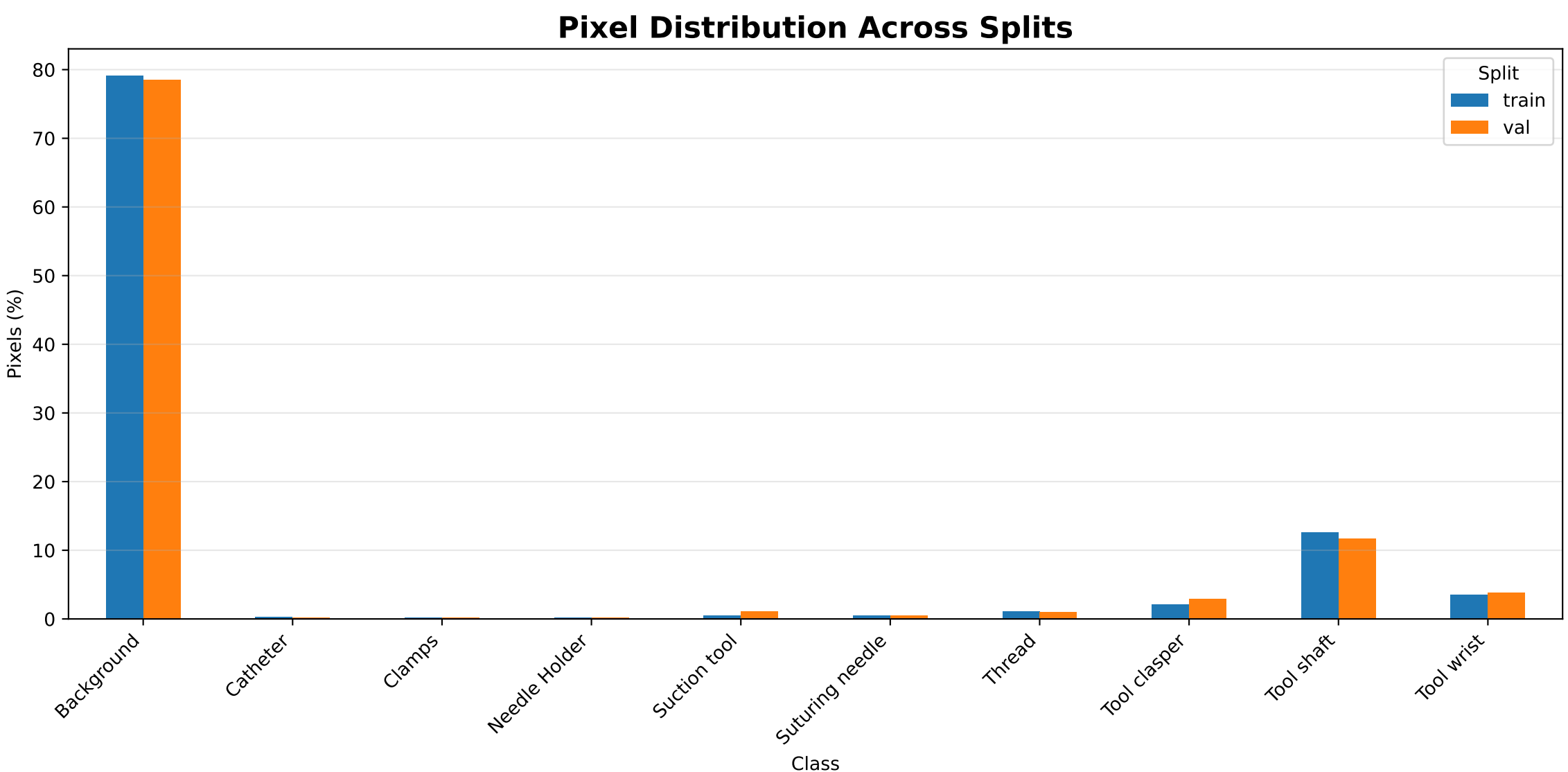
3. Pixel Distribution

Split: train

Class	Pixels	Pixel distribution
Background	21,385,467,795	79.0708%
Tool clasper	580,043,002	2.1447%
Tool wrist	951,629,304	3.5186%
Tool shaft	3,409,308,790	12.6056%
Suturing needle	123,939,049	0.4583%
Thread	297,687,870	1.1007%
Suction tool	138,590,373	0.5124%
Needle Holder	48,120,435	0.1779%
Clamps	48,676,302	0.1800%
Catheter	62,501,880	0.2311%

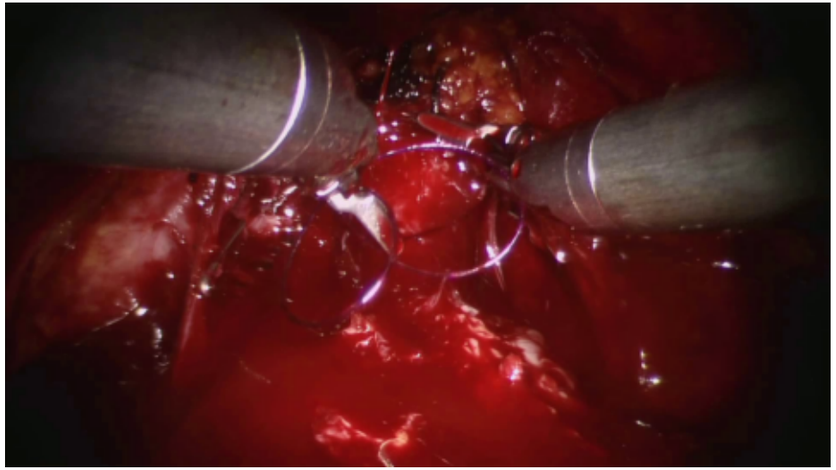



Class Presence Across Splits
Split
train
val
Samples containing class (%)
0
20
40
60
80
100
Background
Catheter
Clamps
Needle Holder
Suction tool
Suturing needle
Thread
Tool clasper
Tool shaft
Tool wrist
Class


Pixel Distribution Across Splits
Split
train
val
Pixels (%)
0
10
20
30
40
50
60
70
80
Background
Catheter
Clamps
Needle Holder
Suction tool
Suturing needle
Thread
Tool clasper
Tool shaft
Tool wrist
Class

4. Qualitative Analysis — val — Sample 2514

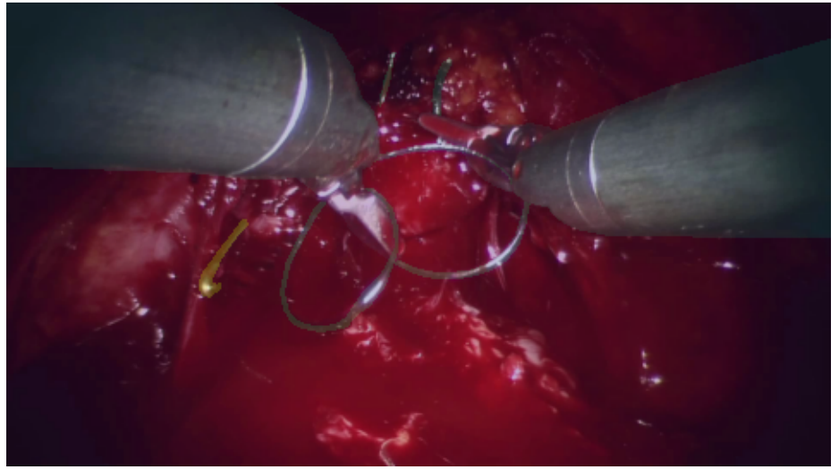
Reference Image



Ground Truth



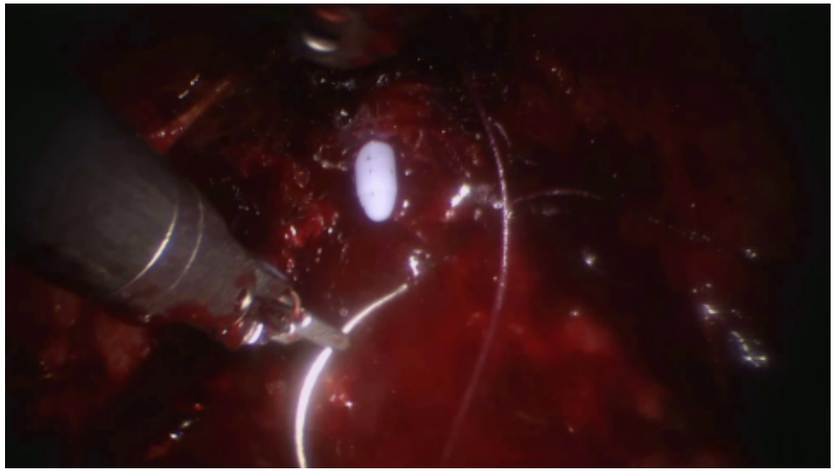
Ground Truth Overlay



Present classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Clamps

4. Qualitative Analysis — val — Sample 1426

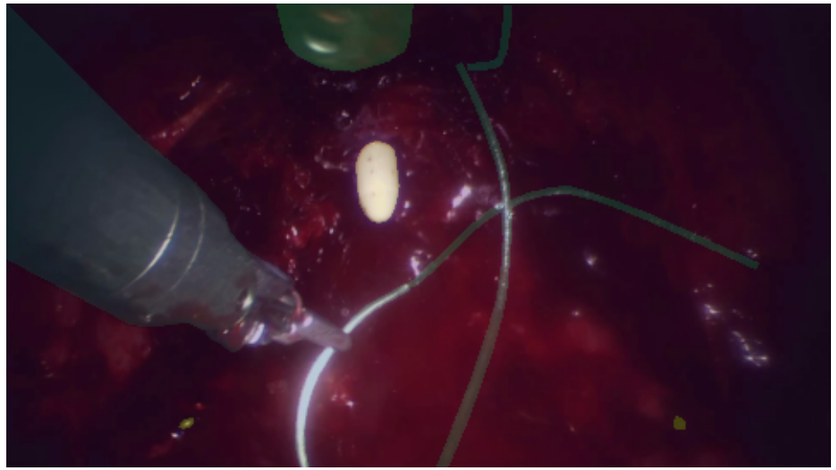
Reference Image



Ground Truth



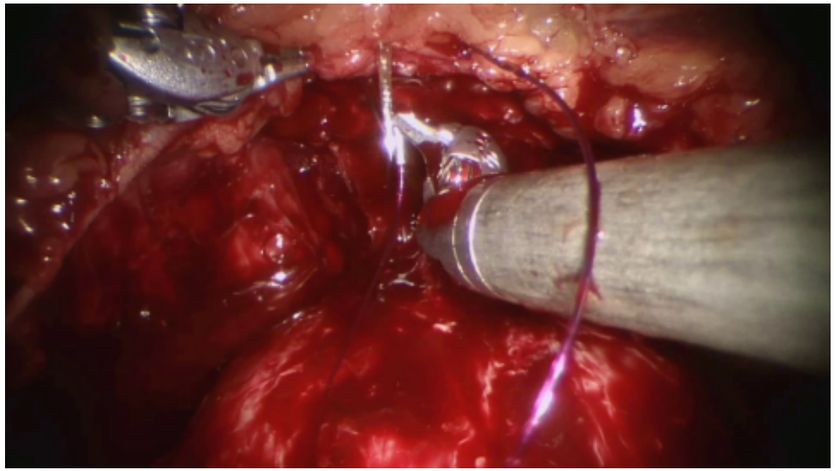
Ground Truth Overlay



Present classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Suction tool, Clamps, Catheter

4. Qualitative Analysis — val — Sample 2127

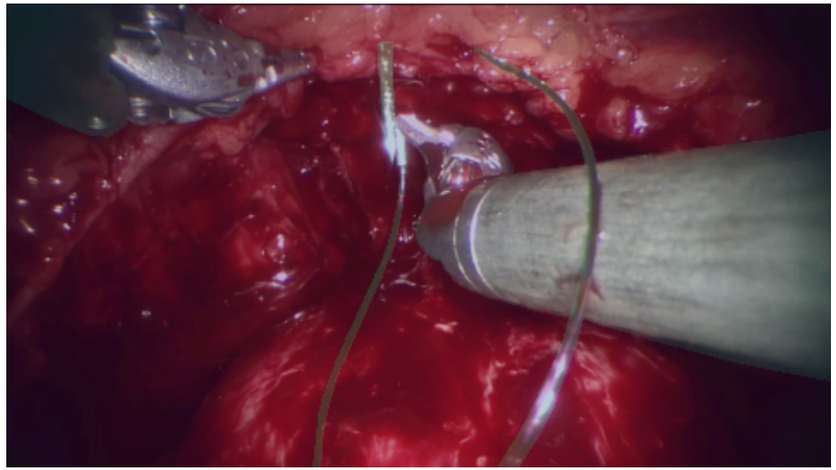
Reference Image



Ground Truth



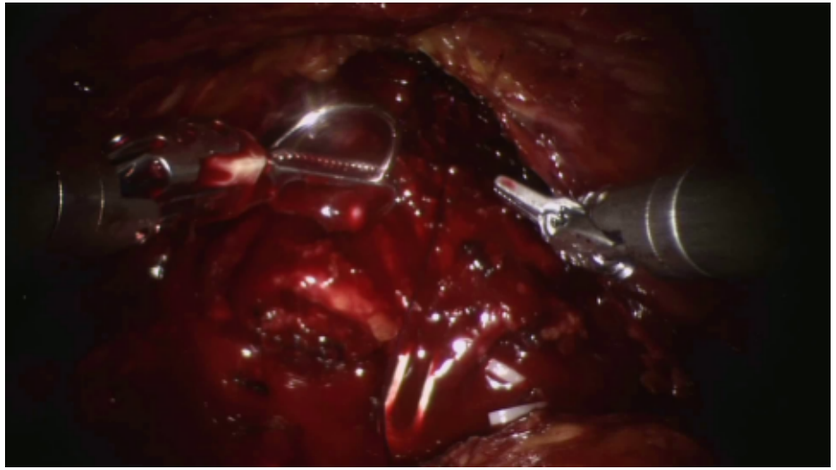
Ground Truth Overlay



Present classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread

4. Qualitative Analysis — val — Sample 289

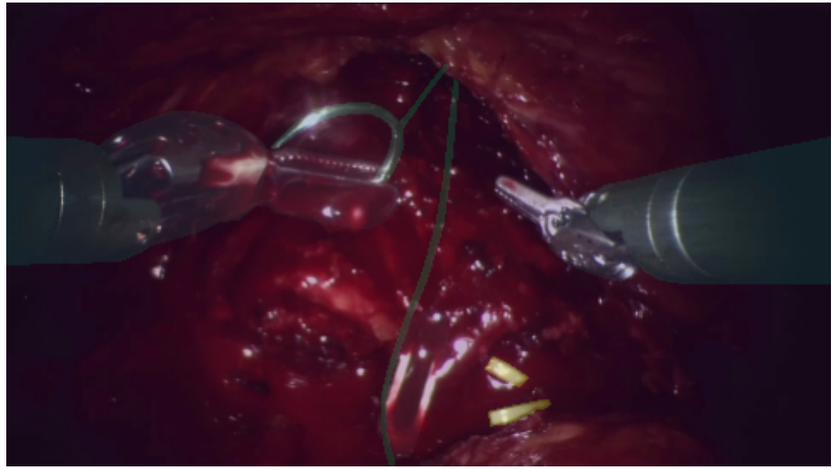
Reference Image



Ground Truth



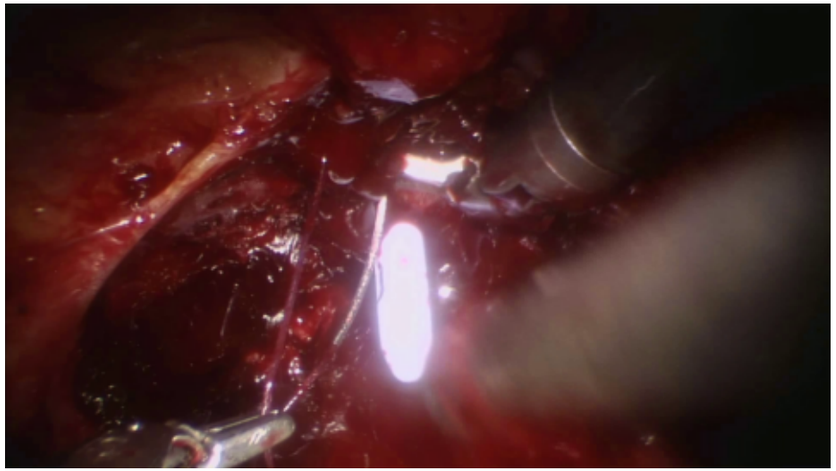
Ground Truth Overlay



Present classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Clamps

4. Qualitative Analysis — val — Sample 1408

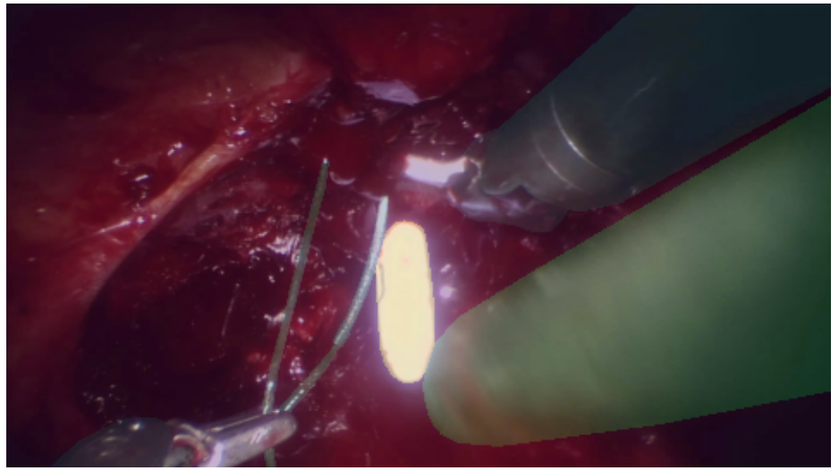
Reference Image



Ground Truth



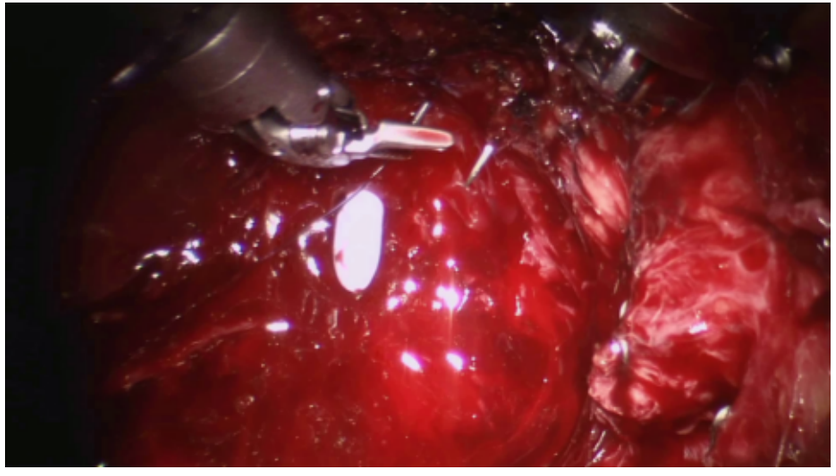
Ground Truth Overlay



Present classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Suction tool, Catheter

4. Qualitative Analysis — train — Sample 13344

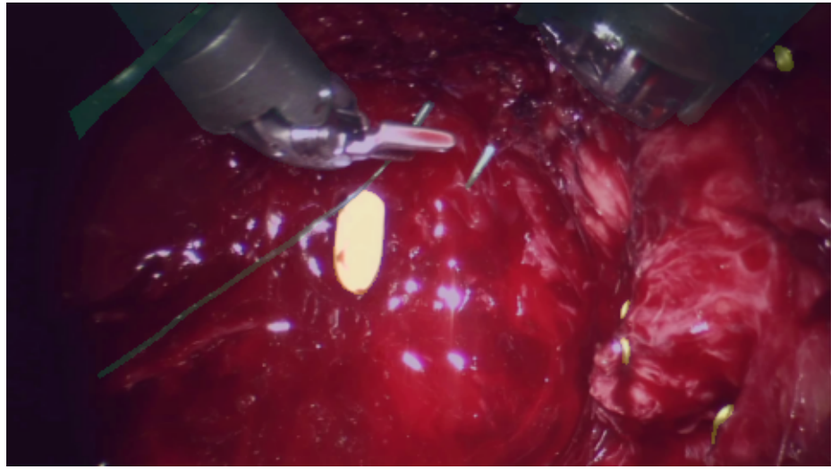
Reference Image



Ground Truth



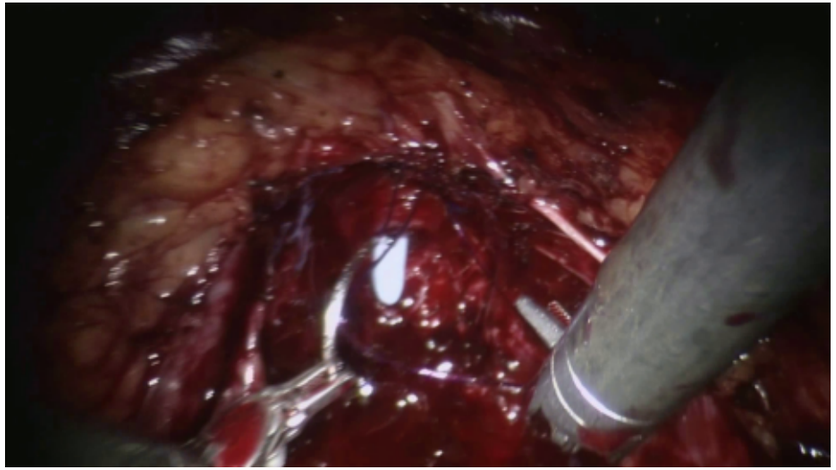
Ground Truth Overlay



Present classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Clamps, Catheter

4. Qualitative Analysis — train — Sample 8975

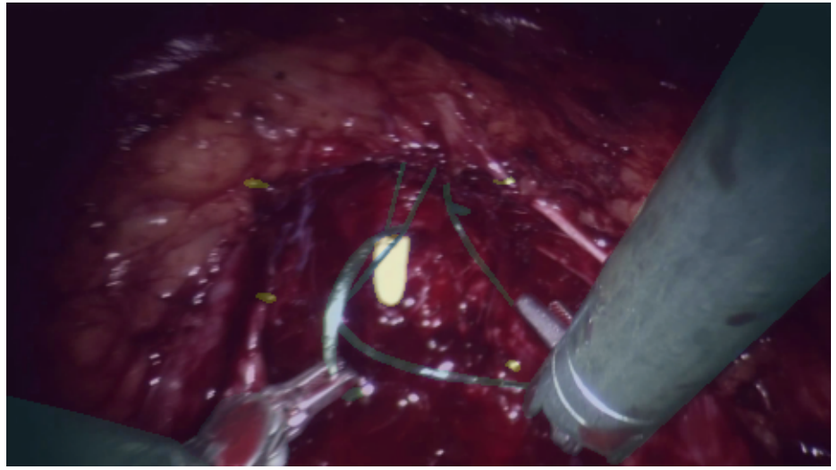
Reference Image



Ground Truth



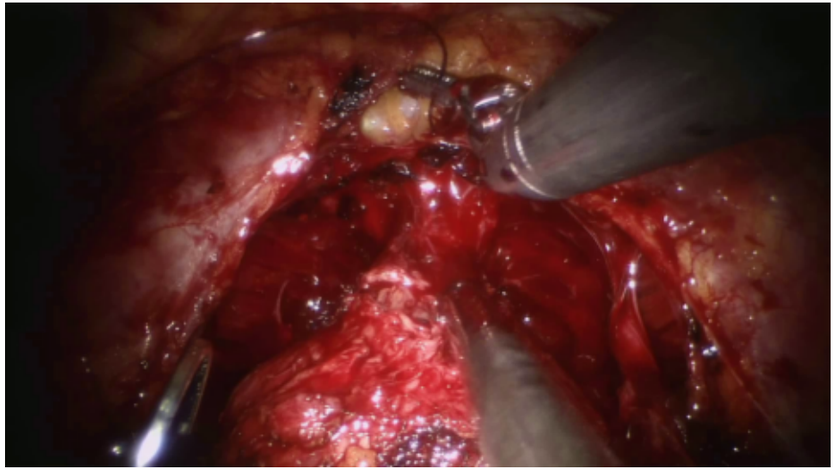
Ground Truth Overlay



Present classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Clamps, Catheter

4. Qualitative Analysis — train — Sample 11788

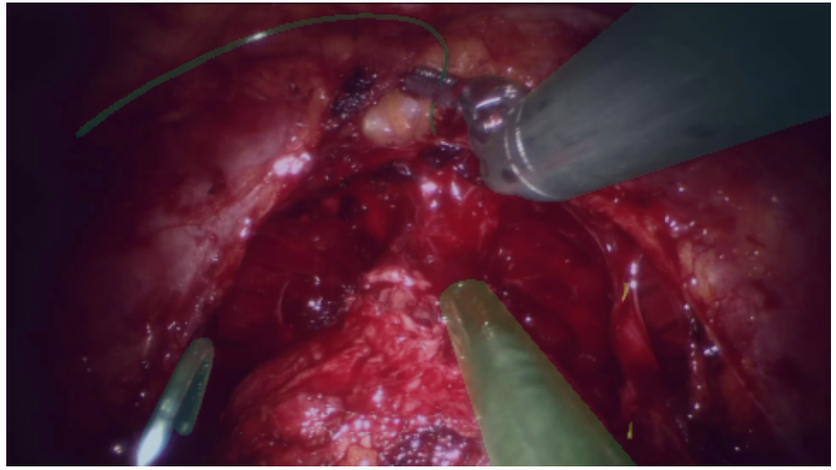
Reference Image



Ground Truth



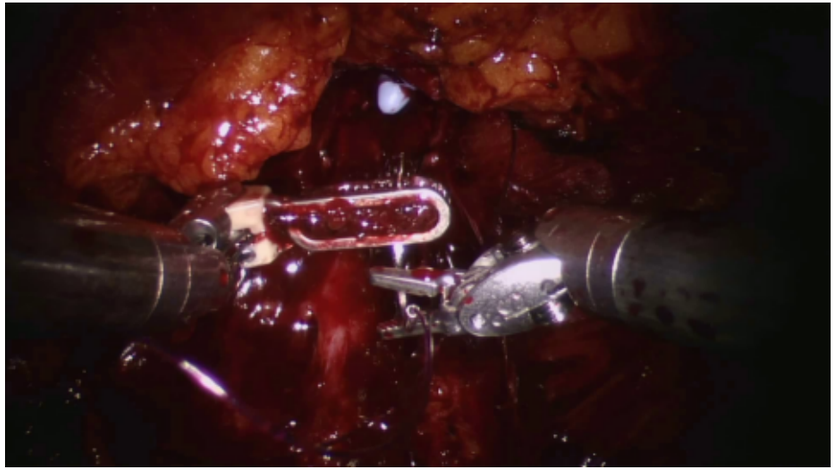
Ground Truth Overlay



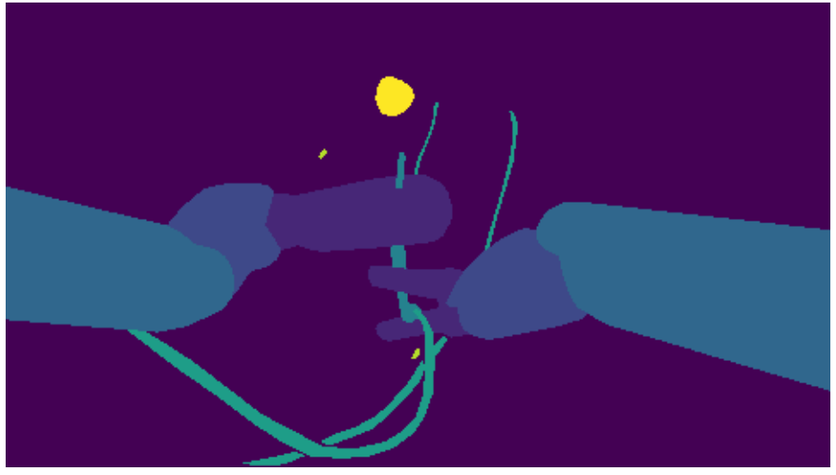
Present classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Suction tool, Clamps

4. Qualitative Analysis — train — Sample 4415

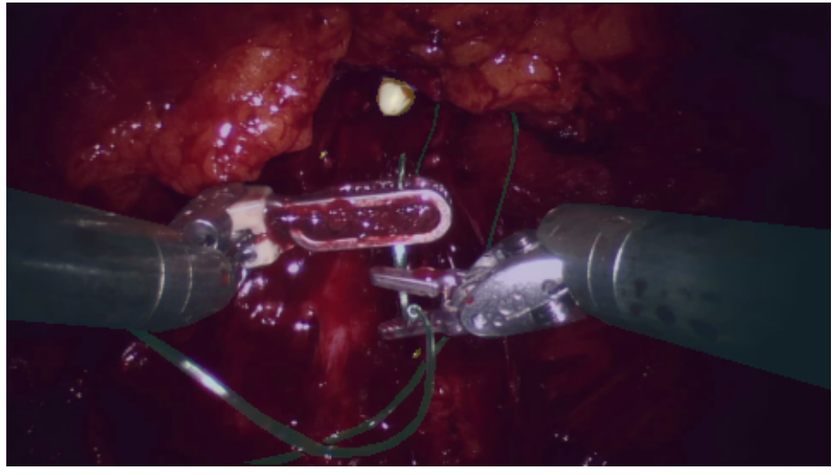
Reference Image



Ground Truth

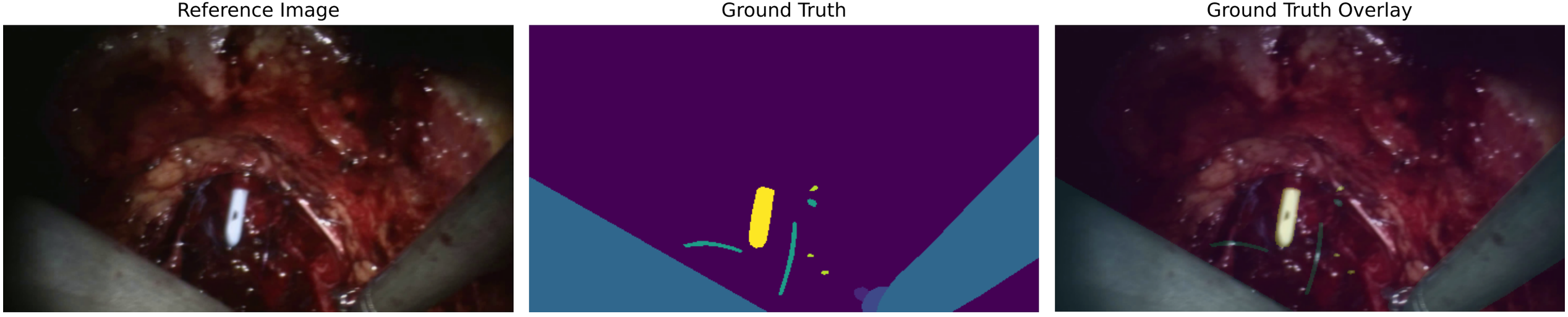


Ground Truth Overlay



Present classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Clamps, Catheter

4. Qualitative Analysis — train — Sample 8899



Present classes: Background, Tool clasper, Tool wrist, Tool shaft, Thread, Clamps, Catheter

5. Validation Summary

Dataset: sarrarp50

Number of classes: 10

Split sizes:

- val: 3,252 samples
- train: 13,043 samples

Class presence:

- val: Needle Holder has the lowest sample presence (2.37%).
- train: Needle Holder has the lowest sample presence (2.92%).

Pixel distribution:

- val: Background is the dominant class (78.49%), while Clamps has the smallest pixel share (0.1655%).
- train: Background is the dominant class (79.07%), while Needle Holder has the smallest pixel share (0.1779%).

Interpretation:

- Sample presence measures how frequently a class occurs across images.
- Pixel distribution measures how much image area is occupied by each class.
- These two statistics describe different forms of class imbalance.
- Differences between splits should be considered when interpreting model performance.
- Group-level stratification cannot guarantee that every rare class appears in every split.